

Portfolio Project 2:

Manufacturing Defects Analysis – ANOVA Approach

Tyrone Muhammad, BSDA, CPC

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Manufacturing Defect Analysis – Electronic Board Production

Introduction

This report presents the findings of a hypothesis test conducted to determine if there is a statistically significant difference in defect rates among three production lines in a manufacturing company located in Tijuana, Mexico. The company specializes in building electronic boards, which go through a welding process in the Manual Finish area. Recent demand increases have led to a rise in welding defects, impacting production efficiency and compliance with IPC-A-610E standards. The analysis was performed using a one-way ANOVA, followed by a Tukey's Honest Significant Difference (HSD) test to identify specific differences between production lines.

Project Case Study

The company manufactures electronic boards, which must go through a welding process that involves:

1. Placement of Thru-Hole components on the boards.
2. Wave soldering using fixtures.
3. Quality inspection to check for welding defects.
4. Final assembly, where additional components are installed.

Recently, there has been an increase in welding defects, including:

- Solder bridges
- Missing components
- Damaged/lifted components
- Insufficient/excessive solder

These defects have caused increased assembly and electronic test failures, making post-production corrections cost-prohibitive due to the required skilled labor and production time.

The company aims to comply with IPC-A-610E quality standards, which set criteria for acceptable welding, such as:

- No cavities or imperfections on the solder surface.
- Good wet conditions between terminals and tracks.
- 100% welding filament around terminals.
- Smooth finish with thin edges on conductor tracks.
- No evidence of raised filaments.

Project Goals

To address these issues, the company has set two key objectives:

1. Reduce defects by 20% in the welding process across the three production lines in the Manual Finish area.
2. Increase production capacity by 20% without raising the defect percentage

Hypothesis Statements

- Null Hypothesis (H_0): There is no significant difference in the defect percentages between the three production lines.
- Alternative Hypothesis (H_d): At least one production line has a significantly different defect percentage compared to the others.

ANOVA Test

```
> str(df) # "Model" should be a factor, "Percentage" should be numeric
'data.frame': 15 obs. of 3 variables:
 $ Model      : Factor w/ 3 levels "Model1","Model2",...: 1 1 1 1 1 2 2 2 2 2 ...
 $ Defects    : int 10 11 12 13 14 10 11 12 13 14 ...
 $ Percentage: num 30 14 11.5 8 5 6.67 3.11 2.56 1.78 1.11 ...
> # Run the ANOVA test
> anova_model <- aov(Percentage ~ Model, data=df)
>
> # Display the ANOVA summary table
> summary(anova_model)
      Df Sum Sq Mean Sq F value Pr(>F)
Model    2   369.6   184.78    5.285 0.0226 *
Residuals 12   419.5    34.96
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

1. The dataset DATAAnova2.csv was imported into R and examined to ensure proper structure.
2. The dependent variable, Percentage (defect rate), was analyzed across the independent variable, Model (production lines).
3. A one-way ANOVA test was conducted at a 0.05 significance level to determine if there were significant differences in defect rates between production lines.

Post Hoc Analysis: Tukey's HSD Test

```
> tukey_results <- TukeyHSD(anova_model)
> print(tukey_results)
Tukey multiple comparisons of means
 95% family-wise confidence level

Fit: aov(formula = Percentage ~ Model, data = df)

$Model
      diff      lwr      upr    p adj
Model2-Model1 -10.654 -20.630789 -0.677211 0.0362961
Model3-Model1 -10.400 -20.376789 -0.423211 0.0409467
Model3-Model2  0.254  -9.722789 10.230789 0.9974603

> save.image("\\\\apporto.com\\dfs\\SNHU\\Users\\tyronemuhamma_snhu\\Documents\\TM_DAT_475\\TMuhammad_DAT475_5_1")
> |
```

Since the ANOVA test confirmed significant differences, a Tukey's HSD test was performed to determine which production lines differed.

- Model1 has significantly higher defect rates than both Model2 and Model3 (p-values of 0.0363 and 0.0409 respectively).
- No significant difference between Model2 and Model3 ($p = 0.9975$).

Conclusion and Recommendations

Based on the findings:

- Model1 has the highest defect rate and should be the primary focus for quality control improvements.
- Model2 and Model3 perform similarly, so their production processes could serve as a benchmark for Model1.
- Further investigation into root causes for Model1's defects is recommended, including:
 - Machine maintenance checks.
 - Operator training improvements.
 - Raw material quality assessments.

By addressing these issues, defect rates across production lines can be optimized, improving overall manufacturing efficiency.